

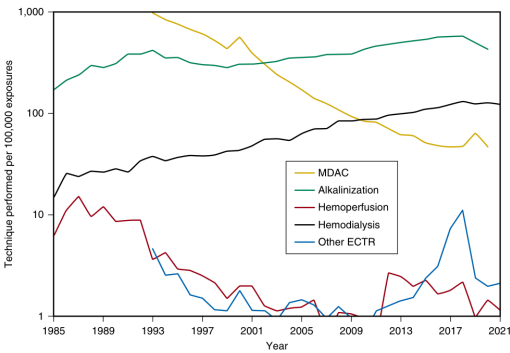
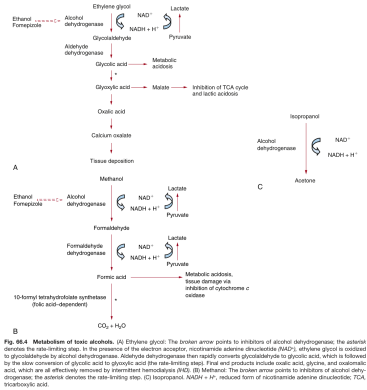


Fig. 66.1 Frequency of various elimination enhancement techniques used in the United States from 1983 to 2021. ECTR, Extracorporeal treatment; MDAC, multiple-dose activated charcoal.

Fig_66_1



Fig_66_4

Table 66.3 Poisons Whose Elimination may be Enhanced by Corporeal Techniques			
Urine Alkalinization	Multiple-Dose Activated Charcoal	Sodium Polystyrene Sulfonate	Prussian Blue
Chlorophenoxy herbicides (2,4-D, MCPA, MCPP) ^a	Amatoxins (Amanita phalloides)	Lithium Potassium	Radiocesium Thallium
Chlorpropamide	Carbamazepine		
Diffenhydramine	Colchicine		
Fluoride	Dapsone		
Methotrexate	Digoxin		
Phenobarbital	Phenobarbital		
Salicylates	Phenytoin		
	Quinine		
	Salicylates		
	Theophylline		
	Yellow cleander		

^a2,4-D, 2,4-dichlorophenoxyacetic acid; MCPA, 4-chloro-2-methylphenoxyacetic acid; MCPP, methylchlorophenoxypropionic acid (Mecoprop).

Table_66_3

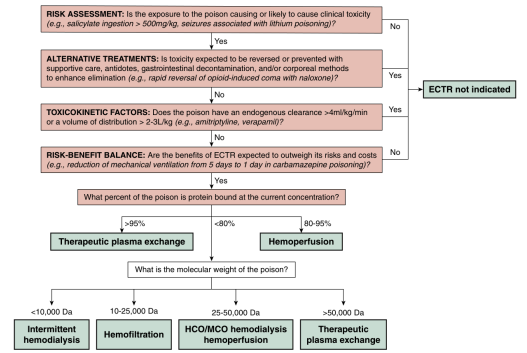


Fig. 66.2 Practical approach for using an extracorporeal treatment in poisonings. This assessment requires analysis of the patient's condition, the exposure, alternative treatments, molecular and toxicokinetic characteristics, and available modalities. ECTR, Extracorporeal treatment; HCO, high cutoff; MCO, medium cutoff; V_p, volume of distribution.

Fig_66_2

Table 66.1 American Association of Poison Control Centers' Statistics for Exposures of Common Dialyzable Poisons (2021)		
Poison	Total Exposures ^a	Deaths
Acetaminophen	137,021	223
Baclofen	4620	11
Barbiturates	1331	3
Carbamazepine	2194	0
Ethylene glycol	6474	20
Gabapentin	20,887	5
Isopropanol	14,706	1
Lithium	6332	10
Metformin	9198	18
(Biguanides)		
Methanol	2228	5
Paracetamol	132	1
Phenytoin	1369	2
Salicylates	30,587	26
Theophylline	141	2
Valproic acid	6908	2

^aIncludes ingestions and mixed formulations.

Table_66_1

Table 66.4 Summary of Extracorporeal Treatments						
	Process	Molecular Weight Cutoff (Da)	Protein Binding Cutoff	Relative Cost	Complications	Comments
Albumin dialysis	Diffusion, adsorption	<60,000-100,000	<95%	+++	++	Liver replacement support
CRRT	Convection and/or diffusion	<10,000-50,000	<80%	++	+	Correction of uremia and acid-base/ E+ disorders
Exchange transfusion	Centrifugation/ separation, filtration	None	None	++	++	Easier than other ECTRs in neonates
Hemodialysis	Diffusion	<10,000	<80%	+	+	Correction of hemolysis and acid-base/ E+ disorders
Hemofiltration	Convection	<50,000	<80%	++	+	Correction of uremia and acid-base/ E+ disorders
Hemoperfusion	Adsorption	<50,000	<95%	++	+++	Saturation of cartridge requires changes
Peritoneal dialysis	Diffusion	<500-5000	<80%	++	++	Low efficacy
Therapeutic plasma exchange	Centrifugation/ separation, filtration	<1,000,000	None	+++	+++	

All extracorporeal treatments above are less likely to be useful for poisons that have a high V_p or a high endogenous clearance. CRRT, Continuous renal replacement therapy; Da, daltons; E+, electrolyte; ECTR, extracorporeal treatment; ++, low; +, moderate; +++, high; +++++, very high.

Table_66_4

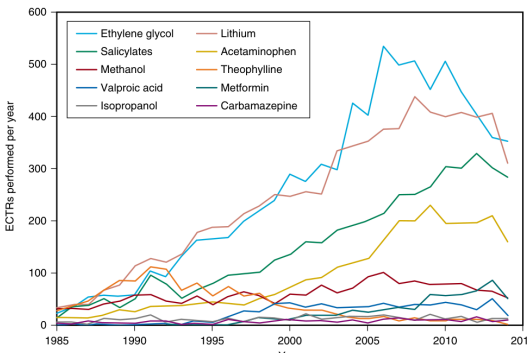


Fig. 66.3 Frequency of extracorporeal treatments used in the United States from 1983 to 2015. ECTR, Extracorporeal treatment.

Fig_66_3

Table 66.2 Common Antidotes Used in Treatment of Poisonings	
Poison	Antidote
Acetaminophen (paracetamol)	N-acetylcysteine
Amanita Phalloides	Silico/ perillidin G
Anticholinergic drugs	Physostigmine for significant sedation
Anticholinesterase insecticides	Atropine and possibly pyridostigmine or edrophonium
Benzocaine	Flumazenil
Calcium channel blockers	Calcium, insulin-dextrose infusion
Cation monoxide	Oxygen
Cyanide	Hydroxocobalamin and/or thiosulfate
Dabigatran	Idarucicab
Digoxin	Digoxin Fab antibody, atropine
Emersonian (e.g., snake and spider)	Antivenom
Ethylene glycol/methanol	Ethanol or fomepizole
Heptaminol	Pharmazone sulfate
Hydroquinone	Deferoxamine
Lead	CaNa ₂ EDTA, or succimer (chelation)
Mercury	DMG or DMSO
Methotrexate	Folinic acid, glucaprase
Nitrofurantoin	Methylene blue
Opioids	Naloxone
Potassium-induced methemoglobinemia (e.g., dapsone, and nitrofurantoin)	Methylene blue
Salicylates	Bicarbonate
Sulfonamides	Oximetazoline, glucose
Tetracycline antibiotics	Bicarbonate
Valproic acid	L-carnitine
Vitamin K	Vitamin K

In the United States, silico/ perillidin G can be obtained by application for an Emergency Investigational New Drug from the FDA during business hours or the FDA Emergency Call Center after hours.

Table_66_2

Table 66.5 Factors that May Enhance Poison Clearance During Hemodialysis

- Larger surface area of dialysis membrane
- High-flux dialyzer
- High blood and dialysate flows
- Increased ultrafiltration rate (with replacement solution)
- Increased time on dialysis
- Reduced recirculation
- Two dialyzers in series
- Two distinct extracorporeal circuits

Table_66_5

Table 66.6 Physiochemical Characteristics and Toxicokinetics of Various Poisons

	Molecular Weight (Da)	Protein Binding	Volume of Distribution (L/Kg)	Endogenous Clearance in Healthy Adults (mL/min/kg)
Acetaminophen	151	20%	1	5
Atenolol	266	5%	1.1	2.1
Baclofen	214	30%	0.9	3
Carbamazepine	236	75%	1.2	1.3
Ethylene glycol	62	0%	0.6	1.8
Gabapentin	171	0%	0.9	1.5
Isoniazid	137	10%	0.7	2 (slow acetylator) 7 (rapid acetylator)
Isopropanol	60	0%	0.6	1.2
Lithium	7	0%	0.8	0.4
Metformin	166	5%	5	10
Methanol	32	0%	0.6	0.7
Methotrexate	454	50%	0.8	1.5
Paraquat	186	5%	1.0	8
Phenobarbital	232	40%	0.7	0.2
Phenytoin	252	90%	0.6	0.4
Pregabalin	159	0%	0.5	1.1
Salicylic acid	138	80% ^a	0.2	1.5
Sotalol	272	0%	1.4	2
Theophylline	180	60%	0.5	0.7
Valproic acid	144	90% ^a	0.2	0.1

^aProtein binding saturation occurs at high concentrations.
Da, daltons.

Table_66_6

Table 66.7 Conversion of Toxic Alcohol Concentration From mmol/L to mg/dL

Toxic Alcohol	Conversion from mmol/L to mg/dL
Acetone	x 5.81
Diethylene glycol	x 10.61
Ethanol	x 4.61
Ethylene glycol	x 6.21
Isopropanol	x 6.01
Methanol	x 3.20
Propylene glycol	x 7.61

Table_66_7

Table 66.8 Antidote Dosage During Toxic Alcohol Poisoning

	40% PO Ethanol (e.g., scotch)	10% IV Ethanol	IV Fomepizole
Loading dose ^a	1.8 mL/kg	7.6 mL/kg	15 mg/kg
Maintenance dose	0.2 mL/kg/h (nondrinker) 0.5 mL/kg/h (chronic drinker)	0.8 mL/kg/h (nondrinker) 2.0 mL/kg/h (chronic drinker)	10 mg/kg q12h x 4 doses, then 15 mg/kg q 12 hours
Maintenance dose during IHD	0.5 mL/kg/h (nondrinker) 0.8 mL/kg/h (chronic drinker)	2.1 mL/kg/h (nondrinker) 3.3 mL/kg/h (chronic drinker)	Same dose but q4h or a constant infusion of 1.0-1.5 mg/kg/h

^aAssumes initial ethanol concentration is zero, dose is independent of chronic drinking status.

Table_66_8